AMRO RABEA

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SUMMARY

AI & Data Science Engineer with hands-on experience delivering high-impact machine learning and deep learning solutions across computer vision, NLP, and large language models (LLMs). Adept at end-to-end project execution from data preprocessing to deployment, achieving measurable improvements in accuracy, efficiency, and usability. Passionate about transforming data into actionable insights and mentoring peers in AI technologies.

EDUCATION

Modern Academy Maadi Cairo, Egypt **Sep 2022 - Present**
Bachelor of Computer Science (B.S.), Computer Science (GPA: 3.34)

EXPERIENCE

Digital Egypt Pioneers Initiative (DEPI) **Nov 2024 - May 2025**
AI & Data Science Fellow *Hybrid*

- Enhanced CNN and traditional ML models, increasing accuracy by 6–12% and reducing inference latency by 30%, preparing models for practical computer vision applications
- Built clustering and analysis pipelines that reduced exploratory data analysis time by 25%, enabling faster model iteration.
- Delivered end-to-end ML workflows using TensorFlow and Scikit-learn, improving team productivity and model reliability.

Scale AI **Sep 2024 - Nov 2024**
Prompt Engineer *Remotely*

- Engineered 100+ structured prompts that improved frontier LLM code-generation accuracy.
- Built Python scripts to automate prompt testing and quality checks, reducing manual review time by 40%.
- Collaborated with engineers using FastAPI + LangChain to resolve model inconsistencies, increasing output stability across edge cases.

Prodigy infotech **Feb 2024 - Apr 2024**
Machine Learning Specialist Intern *Remotely*

- Enhanced a hand-gesture recognition system using MediaPipe + OpenCV, boosting accuracy from 82% to 95% and reducing false detections by 40%.
- Improved a food-image classification model from 78% ’91%, enabling more reliable calorie estimation for health-tracking applications.
- Implemented data augmentation and optimized CNN architectures, resulting in improved training efficiency and reduced overfitting, leading to enhanced model performance

PROJECTS

Autonomous Driving & V2V Communication System **Sep 2025 - Present**

- Developed a lane-keeping assistance system using deep learning + classical CV, achieving 92% lane-detection accuracy in real-world lighting conditions.
- Designed and implemented a lightweight V2V communication protocol enabling vehicles to broadcast speed, location, and direction at 10 Hz with <50 ms latency for cooperative driving.
- Integrated perception, decision-making, and control modules into a unified pipeline, reducing simulated collision risk by 35% through cooperative braking logic.

Intelligent Surveillance System **Jan 2025 - Jul 2025**

- Developed a full-stack surveillance platform enabling semantic video search and real-time object tracking.
- Achieved real-time detection using YOLOv8 and BLIP, integrated with a FastAPI + Streamlit UI.
- Reduced processing time by 40% through asynchronous Celery + Redis pipelines.

Genetic Cancer Prediction & Diagnostic Support Platform **Dec 2024 - Jan 2025**

- Built ML models achieving 94% accuracy in cancer prediction from genomic and imaging data.
- Integrated a medical chatbot powered by RAG for patient-friendly diagnosis explanations.

SKILLS

- **Programming:** C/C++, Python, JavaScript, SQL
- **AI/ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Computer Vision:** OpenCV, YOLO, MediaPipe, Segmentation Models, Image Augmentation
- **MLOps & Deployment:** FastAPI, Docker, Redis, Celery, Streamlit, REST APIs
- **Data Analysis & Visualization:** Power BI, Tableau, Pandas, Matplotlib, Seaborn
- **NLP & LLMs:** LangChain, RAG Pipelines, Prompt Engineering, OpenAI API

AWARDS & ACHIEVEMENTS

- **2nd Place:**Egyptian Collegiate Programming Contest (ECPC) University Round 2024
- **Codeforces Pupil:**+1000 problems solved